DIGITAL TWIN CHALLENGE 2026

VitaTwin

Your Living Digital Health Twin

Cardiometabolic risk · Explainable models · Synthetic data only
Reimagining & Reforming Healthcare in India Summit 2026 · Happiest Health

Team: [YOUR TEAM NAME] · [College / Incubator]

01 · PROBLEM

Reactive care in a rising cardiometabolic burden

India’s burden

Hypertension, type 2 diabetes and early CVD are growing fast; care still reacts after events.

Data is siloed

Wearables and EHR exist, but are rarely fused into one living, trusted patient model.

Black-box risk

Scores without explanation erode trust for patients and clinicians alike.

Privacy barrier

Real PHI is hard to use in student / early PoCs — synthetic-first is the safe path.

A living, explainable digital health twin

1	**Fuse**	Wearable streams + EHR snapshot → one twin state
2	**Score**	CVD 10-yr, diabetes progression, acute stress — with factor contributions
3	**Alert**	Prioritised alerts in plain language + recommended action
4	**Guide**	Patient + clinician views; what-if interventions

Who it serves

Patient / care-seeker

Adults 25–65 at risk of diabetes, hypertension or early CVD. Clear risk signals and daily guidance.

Clinician / care team

Doctors and educators who need a living, explainable patient model they can trust.

Demo twin

52-year-old Indian adult with hypertension, T2DM and hyperlipidaemia — aligned to national burden.

Transparent engineering, explainable risk

TECHNICAL STACK

- Next.js 15 (App Router) + TypeScript
- Tailwind CSS — light theme only
- Supabase (Postgres + Auth + RLS)
- Recharts + custom twin SVG
- Vitest + Playwright
- Vercel deploy

AI / RISK MODELS

- 10-yr CVD — factor-weighted, explainable
- Diabetes progression — HbA1c + lifestyle
- Acute cardiac stress — HR/HRV + sleep
- Server-side computation only
- What-if re-runs same validated models
- Not a medical device / not a diagnosis

Browser never holds secrets

UI (Next.js + Tailwind light)

Landing, login, dashboard, vitals, insights, clinician, timeline

Server actions / twin engine

Fusion · risk models · alerts · recommendations · what-if

Data layer

Synthetic generators (offline demo) + Supabase Postgres + Auth + RLS

What the jury can verify

End-to-end twin journey

Landing → demo twin → dashboard → vitals → insights → clinician

Explainable risk

Every score shows contributing factors — no black box

What-if simulator

Change sleep / activity / adherence and see risk deltas

Security baseline

RLS-ready auth, rate limit, CSP/HSTS headers, scrubbed errors

Quality gates

42 unit tests, typecheck, lint, production build

Privacy stance

Synthetic data only — clear banners throughout

Thank you

VitaTwin — living, explainable, synthetic-first digital health twin

Repo: github.com/Pavan3030-pr/VitaTwin
Branch: feat/jury-upgrade · License: MIT
All patient data is synthetic